

Salesforce Developer Bridge Program

Day 5 – Self Learning Track Report

Building Complete Business Transactions with SOQL, DML and Apex

Project: Placement Management System

Student: Sudharshan Kuppili

PIN: 24PA5A0517

Student Details

Field	Details
Name	Sudharshan Kuppili
PIN No.	24PA5A0517
Project	Placement Management System
Day	Day 5 – Retrieving and Managing Information with SOQL and DML
Technology	Apex, SOQL, DML, Salesforce

1. Day Objective

The objective of Day 5 was to strengthen Apex development by retrieving Salesforce records with SOQL, validating business requirements, creating and updating records with DML, preventing duplicate applications, and completing a full Placement Application business transaction.

Core Sequence

```
Retrieve information -> Validate business rules -> Perform DML -> Confirm result
```

2. Business Scenario

The Placement Management System must allow a student to apply for a job only when the business requirements are satisfied.

The system retrieves Student details, Job eligibility requirements, and existing applications before making a decision. Business checks include:

- Duplicate application
- Minimum CGPA
- Eligible Branch
- Active Backlog Limit
- Application Deadline

Only after all validations pass is the Placement Application created.

3. Architecture / Transaction Flow

Apex service class: ApplicationService

Responsibilities:

- Retrieve Student information
- Retrieve Job information

- Check duplicate applications
- Validate eligibility
- Create Placement Application records
- Update Application Status
- Record Interview Date when status becomes Interview Scheduled

Complete Business Transaction Flow

```
Receive Request
-> Retrieve Student
-> Retrieve Job
-> Check Duplicate
-> Validate Eligibility
-> Create Application
-> Save Record
-> Display Confirmation
```

4. Student Information Retrieval

Purpose

Before validating an application, the software must know who the student is. The service retrieves only the fields required for eligibility validation, avoiding unnecessary data retrieval.

Implementation

The submitApplication(Id studentId, Id jobId) method begins by retrieving the required Student fields using SOQL: Id, Name, CGPA__c, Branch__c, Active_Backlogs__c, and Graduation_Year__c.

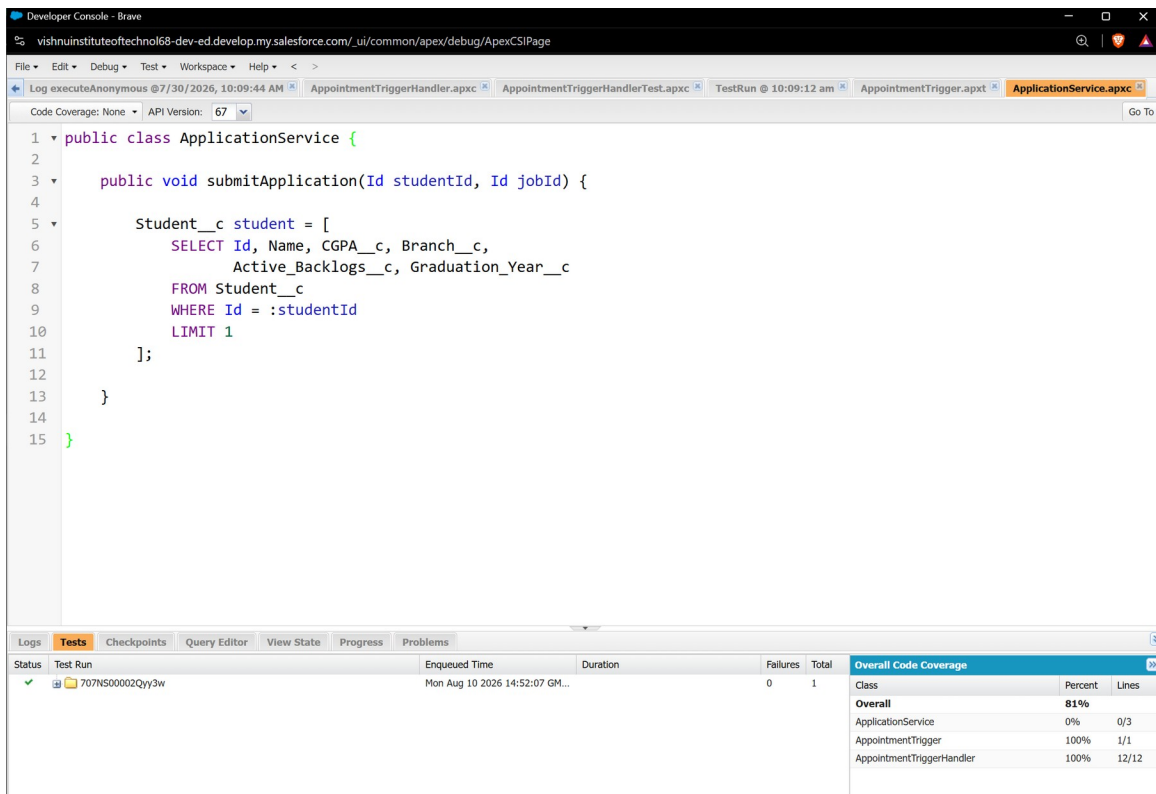


Figure 1: ApplicationService.submitApplication() — Student SOQL retrieval

Observed Result

The application successfully identifies the student before any validation begins.

5. Job Information and Eligibility

Purpose

Every company defines its own eligibility criteria. The software must retrieve those requirements before making a decision.

Implementation

The method was extended to retrieve the selected Job's fields using SOQL: Id, Name, Company_Name__c, Minimum_CGPA__c, Eligible_Branch__c, Active_Backlog_Limit__c, and Application_Deadline__c.

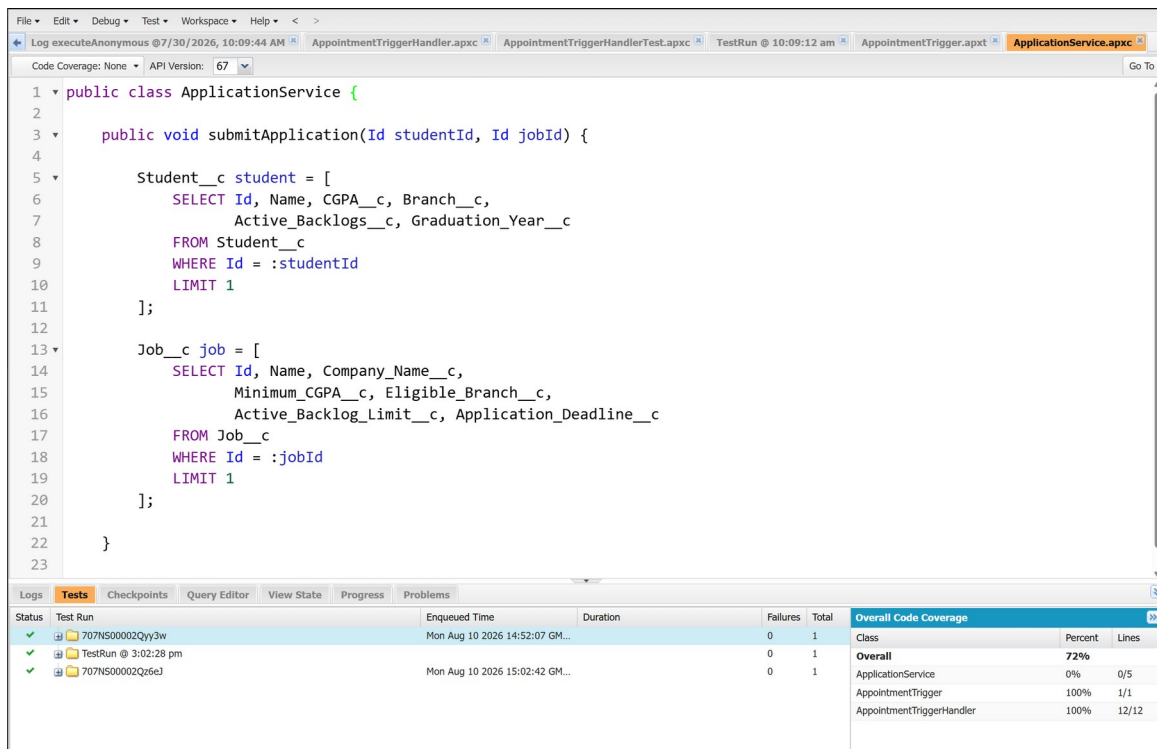


Figure 2: `ApplicationService.submitApplication()` — Student SOQL and Job SOQL retrieval together

Observed Result

The application now possesses all the information necessary for the eligibility decision.

6. Duplicate Application Prevention

Purpose

Students should not accidentally submit multiple applications for the same job. This check runs before eligibility validation and before any DML.

Implementation

The method retrieves any existing Placement Application for the same Student and Job combination. If a matching record is found, processing stops and no new application is created.

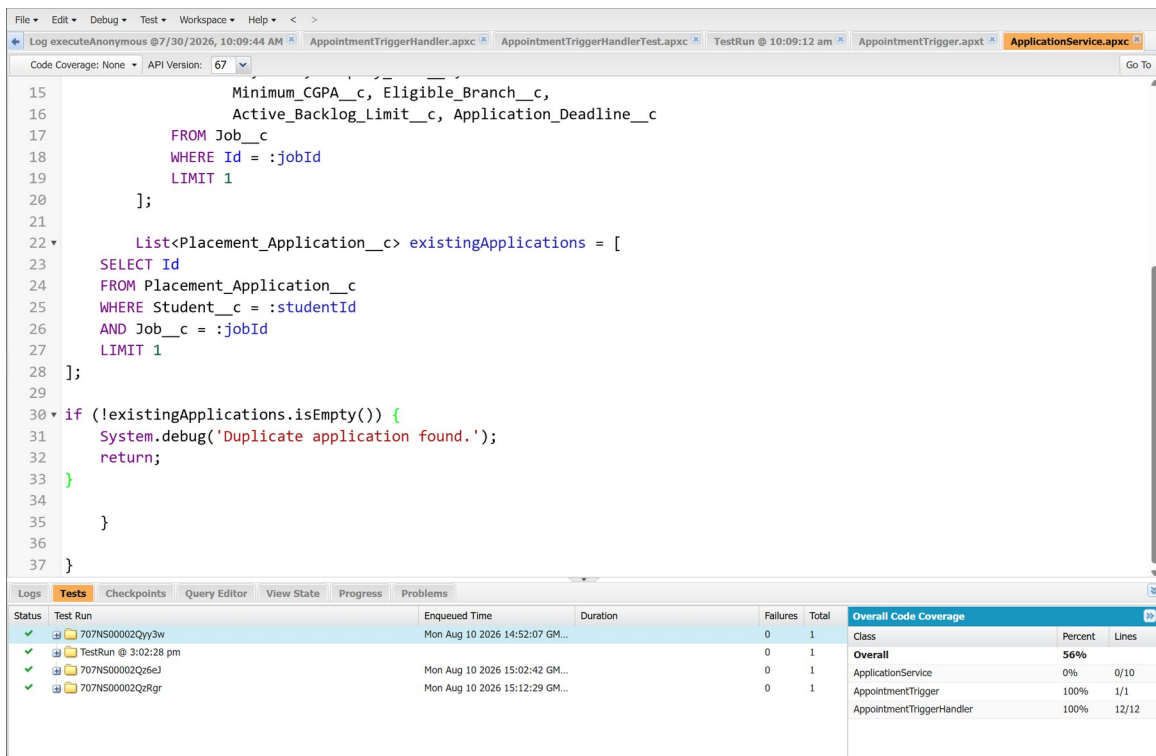


Figure 3: Duplicate check — existingApplications query and early-return logic

Testing

submitApplication() was executed with a Student and Job combination that already had an existing Placement Application. The debug log confirmed the duplicate was correctly detected: "Duplicate application found."

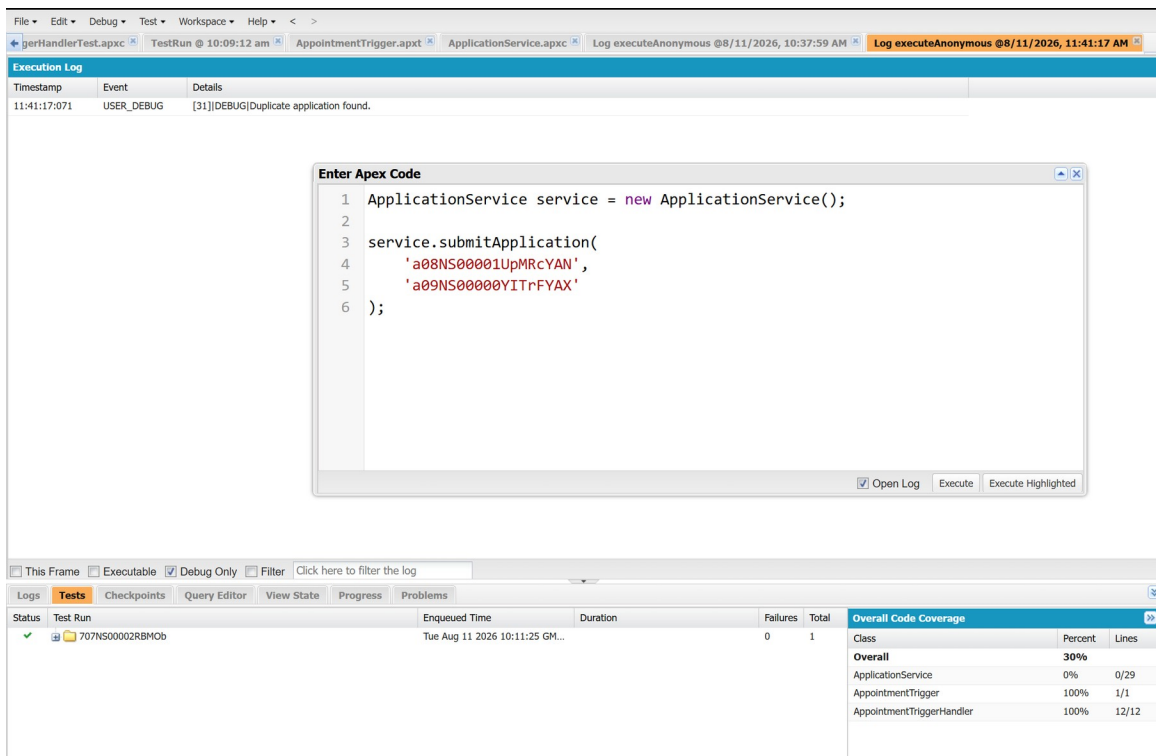


Figure 4: Debug log — "Duplicate application found." and processing stopped

7. Business Validation

The application is rejected when any of the following conditions are true, evaluated in sequence before any DML operation occurs:

- CGPA is below the Job's minimum CGPA
- Branch does not match the Job's eligible branch
- Active Backlogs exceed the allowed limit
- Application Deadline has passed

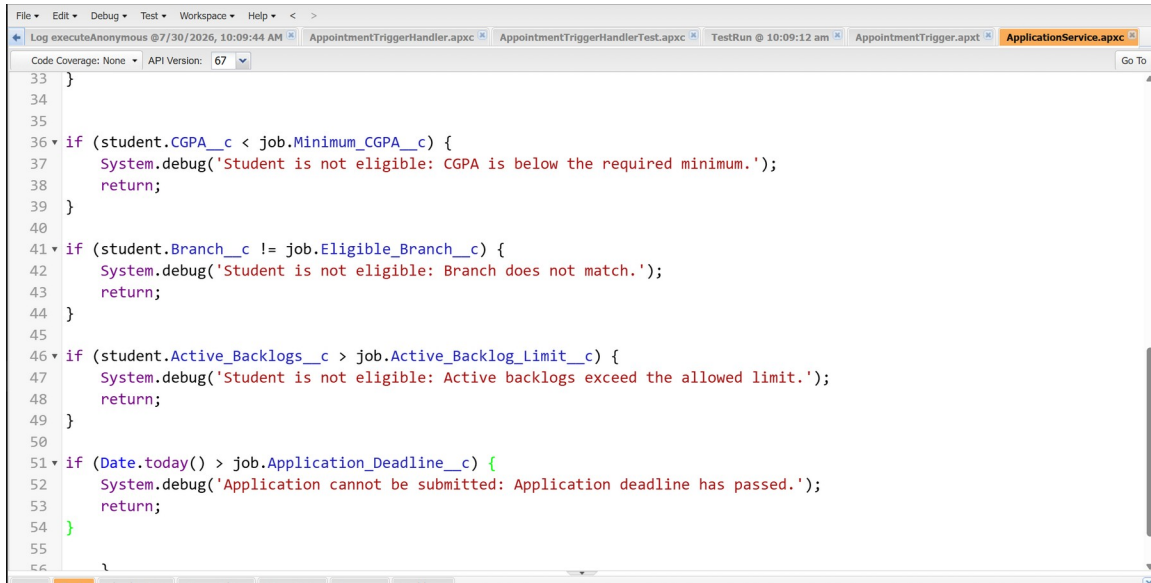


Figure 5: Business validation — CGPA, Branch, Active Backlog, and Deadline checks, each with an early return

Each failed condition returns immediately with a descriptive debug message, ensuring the record is never saved unless every business rule has been satisfied.

8. Creating the Placement Application

Purpose

After all validations succeed, a Placement_Application__c record is created and saved using DML.

Implementation

Fields populated: Student, Job, Application Date, and Status = Applied. The record is saved using an insert DML statement, followed by a confirmation debug message.

Testing

submitApplication() was executed with a valid Student/Job pair that passed every business rule. The debug log confirmed: "Placement Application created successfully."

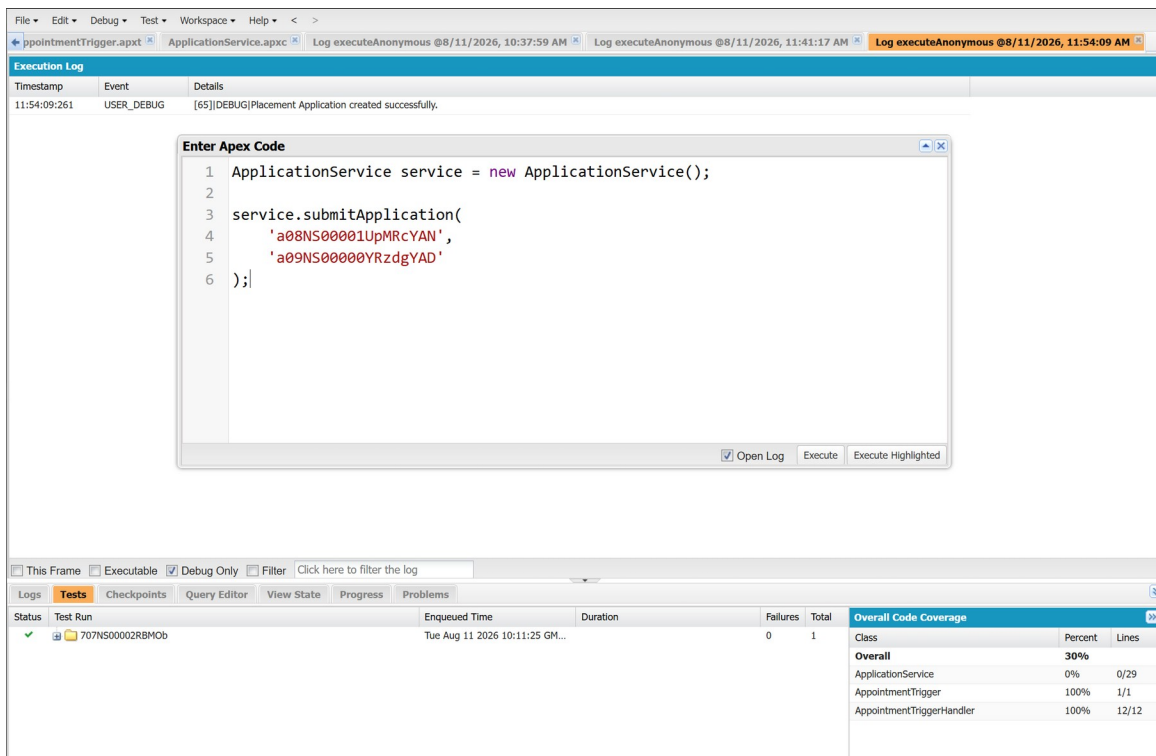


Figure 6: Debug log — "Placement Application created successfully."

The newly created record, PA-0005, was verified directly in Salesforce, linked to the correct Student and Job with Status = Applied.

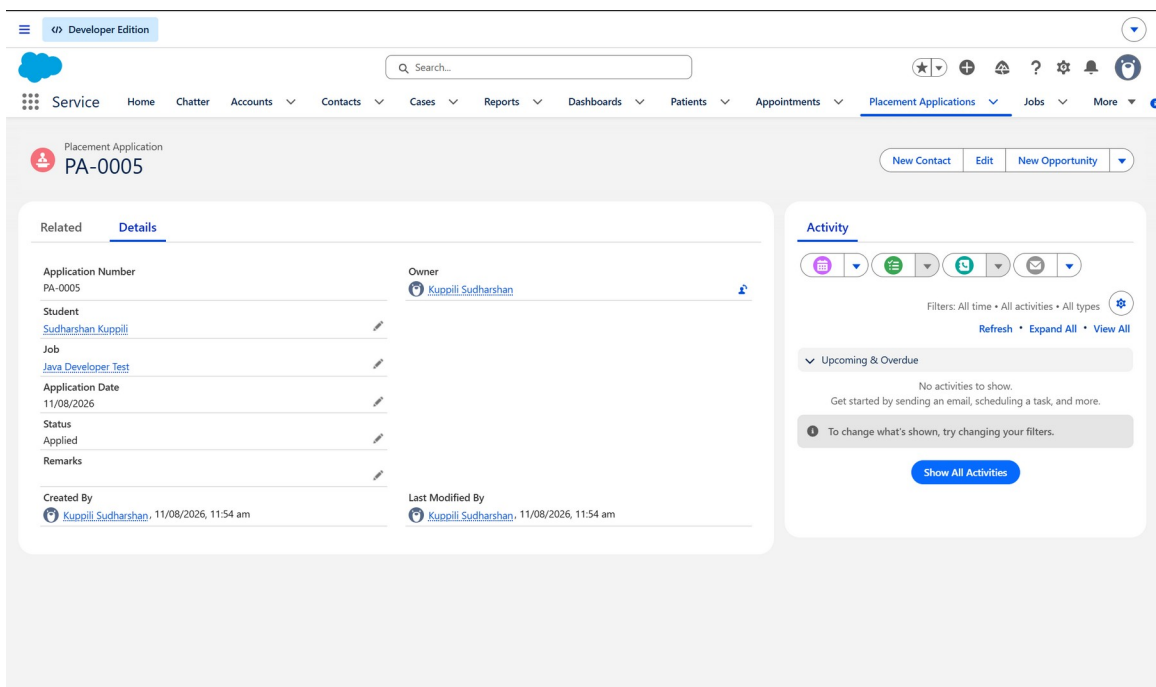


Figure 7: Created Placement Application record PA-0005 — Student, Job, Application Date, and Status = Applied

9. Complete Business Transaction

The successful test above confirmed that the complete transaction sequence worked end-to-end, in this order:

```
Receive Request
-> Retrieve Student
-> Retrieve Job
-> Check Duplicate
-> Validate Eligibility
-> Create Application
-> Save Record
-> Display Confirmation
```

Every step depended on the previous one succeeding, and the transaction only reached the Create Application / Save Record steps once retrieval, duplicate-checking, and validation had all passed.

10. Application Status Update

Purpose

A recruiter finishes interviewing (or reviewing) a candidate and the application status must change. Rather than creating a new record, the existing Placement Application is retrieved and updated.

Implementation

ApplicationService contains `updateApplicationStatus(Id applicationId, String newStatus)`, which retrieves the existing record, updates its Status field, and saves it using DML. Possible statuses include Applied, Shortlisted, Interview Scheduled, Selected, and Rejected.

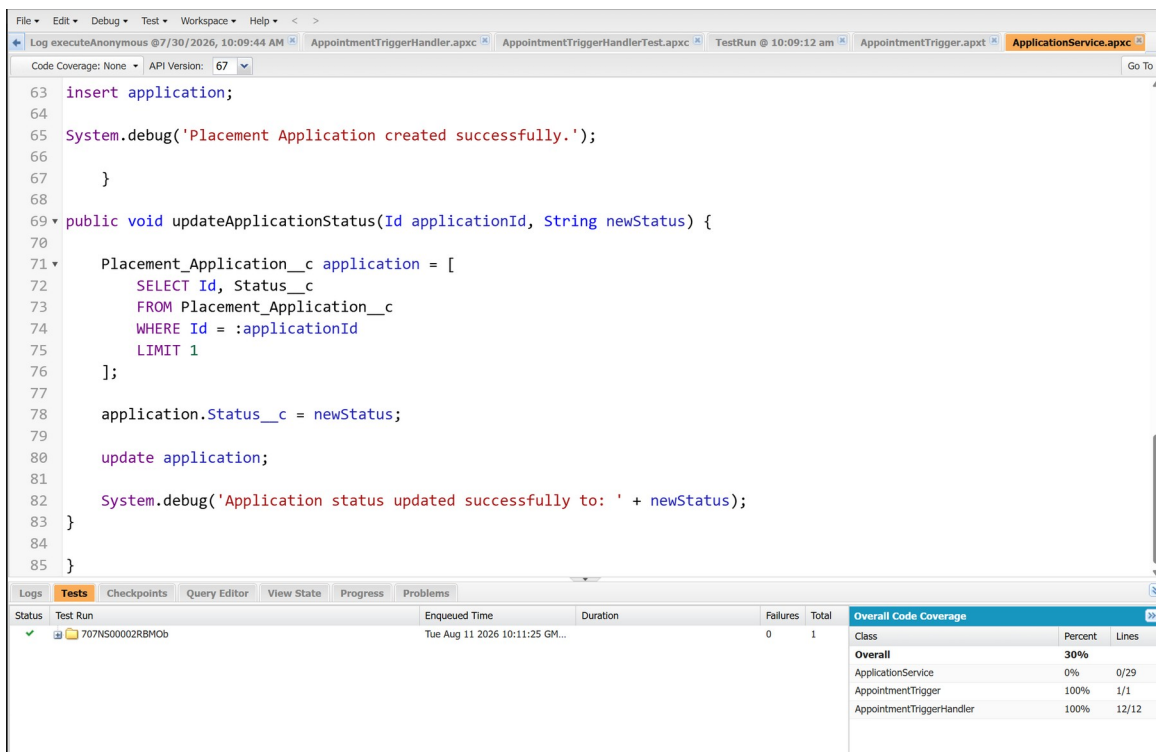


Figure 8: `updateApplicationStatus()` — retrieves, updates Status__c, and saves via DML

Testing

updateApplicationStatus() was executed to change an existing application's status to "Shortlisted". The debug log confirmed the update.

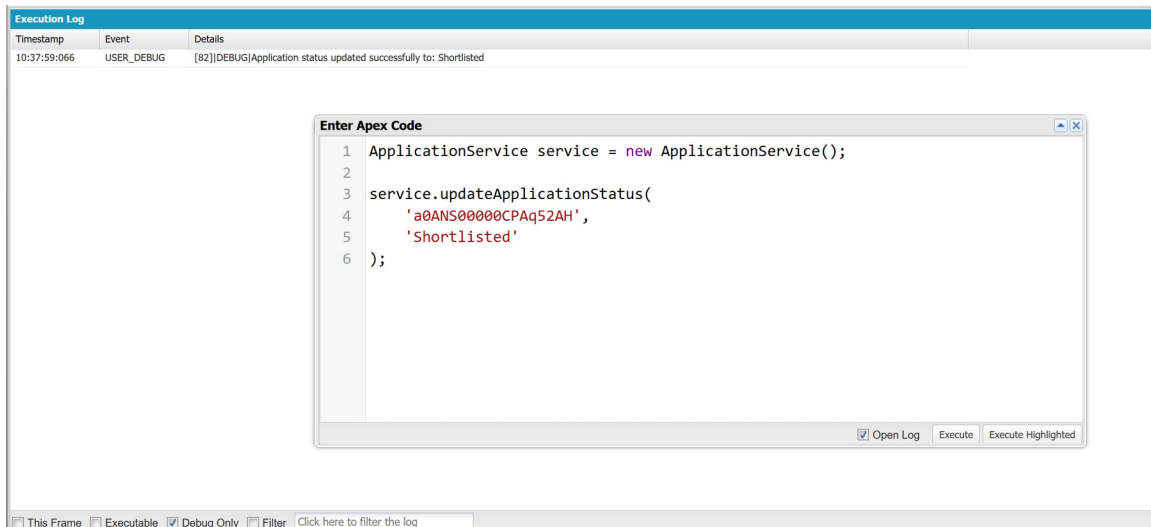


Figure 9: Debug log — "Application status updated successfully to: Shortlisted"

The Placement Application record was verified directly in Salesforce, now showing Status = Shortlisted.

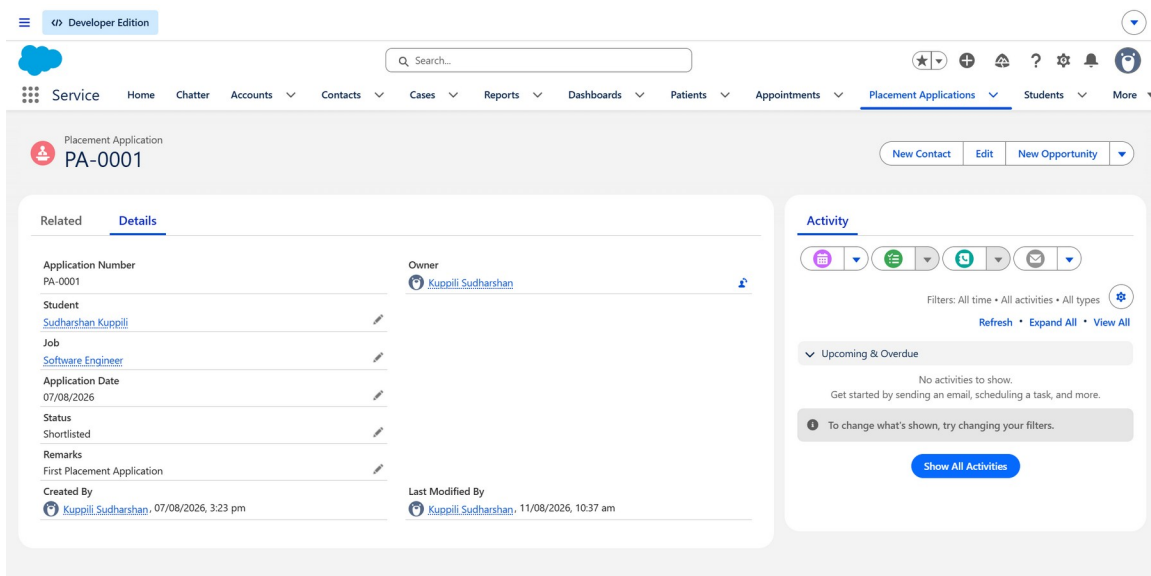


Figure 10: Placement Application PA-0001 — Status updated to Shortlisted

11. Interview Date Enhancement

A Sprint 11 enhancement was implemented per the stretch challenge: when Status changes to Interview Scheduled, automatically populate an Interview Date.

Implementation

- Created a new field: Interview_Date__c (Date) on the Placement Application object.
- Updated updateApplicationStatus() to add an Interview Scheduled condition.
- When newStatus == 'Interview Scheduled', Interview_Date__c is set to Date.today().

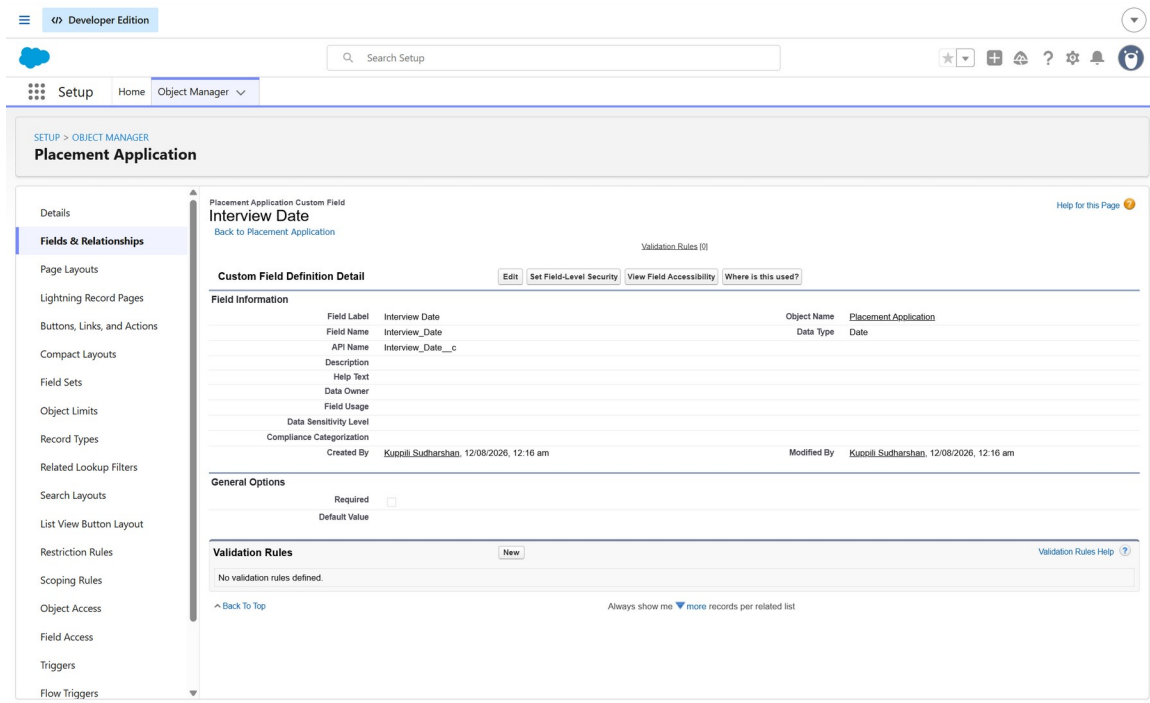


Figure 11: Interview Date custom field (Interview_Date__c) on the Placement Application object

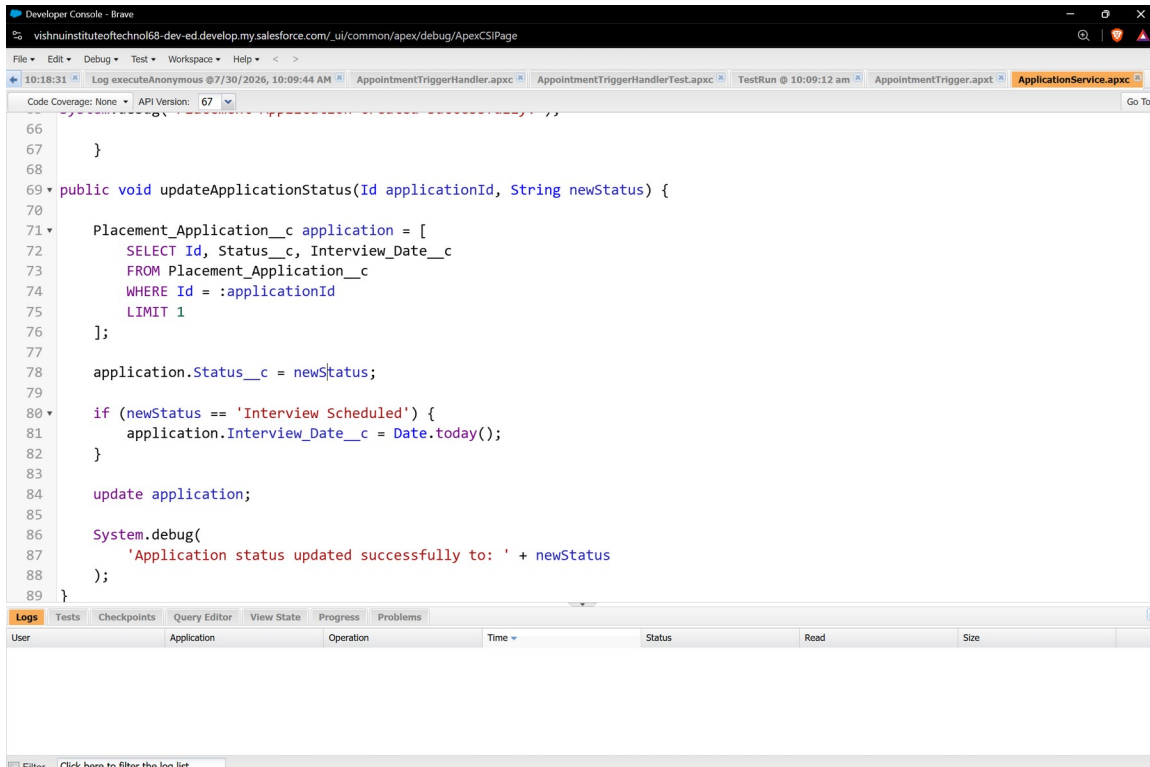


Figure 12: Updated `updateApplicationStatus()` — Interview Scheduled condition sets `Interview_Date__c = Date.today()`

Testing

`updateApplicationStatus()` was executed with `newStatus = "Interview Scheduled"`. The debug log confirmed the update.

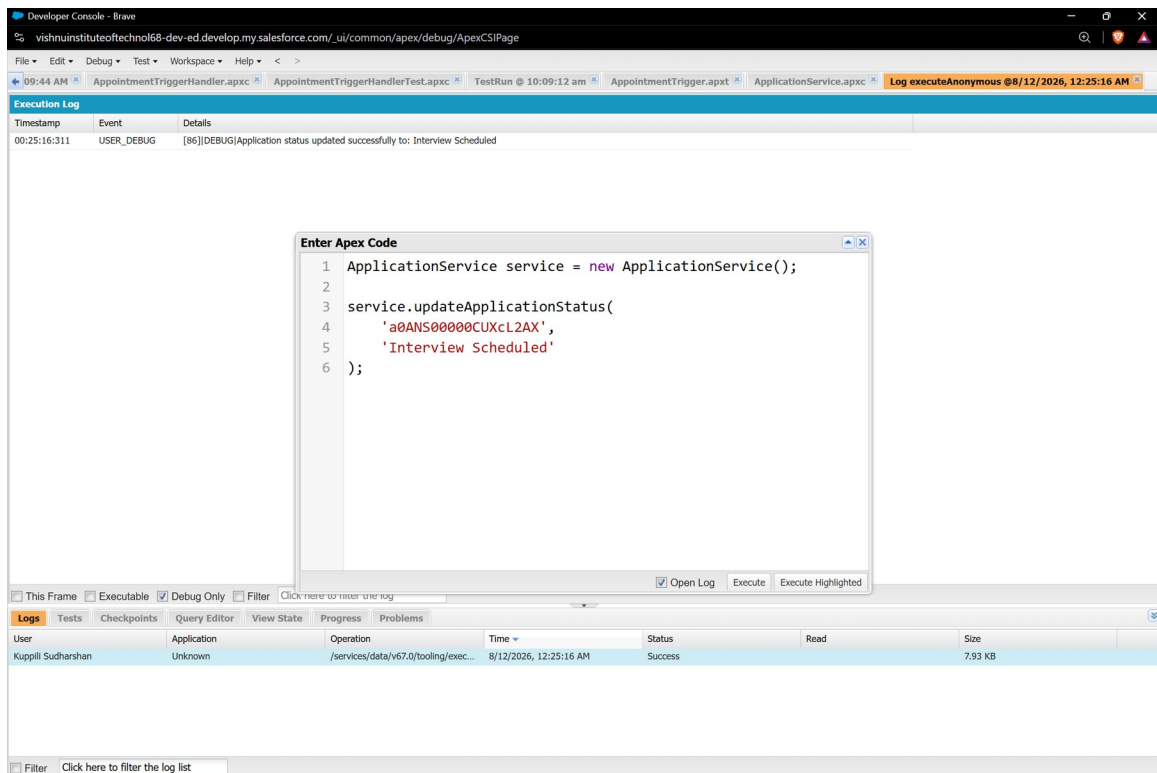


Figure 13: Debug log — "Application status updated successfully to: Interview Scheduled"

The Placement Application record was verified directly in Salesforce, showing Status = Interview Scheduled and Interview Date automatically populated.

Developer Edition

Service Home Chatter Accounts Contacts Cases Reports Dashboards Patients Appoint

Placement Application
PA-0005

Related Details

Application Number PA-0005	Owner Kuppili Sudharshan
Student Sudharshan Kuppili	
Job Java Developer Test	
Application Date 11/08/2026	
Status Interview Scheduled	
Remarks	
Interview Date 12/08/2026	
Created By Kuppili Sudharshan, 11/08/2026, 11:54 am	Last Modified By Kuppili Sudharshan, 12/08/2026, 12:25 am

Figure 14: Placement Application PA-0005 — Status: Interview Scheduled, Interview Date: 12/08/2026

12. Debugging / Engineering Thinking

Key lessons from this sprint:

- A single-record SOQL assignment can throw a QueryException when no record is returned.
- DML can fail because of a Validation Rule or another database constraint.
- Repeated SOQL can cause governor-limit problems.
- Bulk processing should use collections such as List and Map.
- DML exceptions should be handled with try-catch and meaningful error handling.
- Repeated SOQL logic should be centralized or reused where appropriate.
- Business validation must happen before database changes.
- Only required fields should be retrieved.

13. Interview Corner

Why SOQL before DML?

SOQL retrieves the required records and information before business validation and DML.

Why should every SOQL query answer a business question?

Each query should retrieve information required for a specific business decision, avoiding unnecessary data retrieval.

Why avoid unnecessary fields?

Unnecessary fields waste resources and make queries less efficient.

Why validate before DML?

Validation ensures only records satisfying business requirements are saved.

How does service design help another developer?

Clear responsibilities, readable methods, and maintainable structure make the software easier to understand and modify.

14. Sprint Retrospective

What worked well?

The complete Placement Application transaction worked successfully. SOQL, validation, duplicate checking, and DML worked together correctly.

Which part felt most natural?

SOQL retrieval and eligibility validation felt natural because the business rules were clearly defined.

Which business rule was challenging?

Duplicate checking and combining multiple eligibility conditions required careful reasoning.

Was the challenge technical or business-related?

It was a combination, with understanding the business requirement being especially important.

What would be redesigned?

The service could be divided into smaller reusable methods for retrieval, validation, duplicate checking, application creation, notification, and error handling.

How would the system support 10,000 students?

Use bulkified Apex, Lists/Maps, efficient SOQL, avoid SOQL/DML inside loops, and use reusable services that respect governor limits.

"Today I realised that..."

"...retrieving the correct information, validating business rules, and performing DML in the correct order is essential for building reliable and maintainable enterprise software."

15. Test Results

Test Case	Expected Result	Actual Result	Status
Duplicate application check	Processing stops; no new record created	"Duplicate application found." logged; processing stopped	PASS
Valid application creation	Placement Application created	PA-0005 created successfully, Status = Applied	PASS
Status update (Shortlisted)	Existing record updated, not duplicated	PA-0001 updated to Status = Shortlisted	PASS
Interview Date enhancement	Interview Date auto-populated when status = Interview Scheduled	PA-0005 updated to Status = Interview Scheduled with Interview Date = 12/08/2026	PASS

16. Conclusion

Day 5 successfully extended the Placement Management System into a complete, working business transaction. The ApplicationService class now retrieves accurate Student and Job information using SOQL, validates every business rule — duplicate applications, CGPA, branch eligibility, active backlogs, and deadlines — before any data changes, and performs DML responsibly to create and update Placement Application records. The system was further enhanced to automatically record the Interview Date whenever an application's status changes to Interview Scheduled. Together, these pieces demonstrate a real, end-to-end Placement Management business transaction: retrieve, validate, act, and confirm.

Appendix A — Supporting Reference Data

The following records provided the underlying data used across the Day 5 testing above.

The screenshot shows a web application interface for a CRM system. At the top, there's a navigation bar with a search bar and various icons. Below it, a menu bar lists different sections: Service, Home, Chatter, Accounts, Contacts, Cases, Reports, Dashboards, Patients, Appointments, Placement Applications, Students, and More. The 'Students' section is currently selected. The main content area displays a student record for 'Sudharshan Kuppili'. The record includes fields for Student Name, CGPA (8.75), Branch (CSE), Active Backlogs (0), Graduation Year (2027), and Created By (Kuppili Sudharshan, 07/08/2026, 3:14 pm). The 'Details' tab is active, showing the student's information. To the right, there's an 'Activity' section with a filter dropdown set to 'All time' and 'All activities'. Below the filter, it says 'No activities to show. Get started by sending an email, scheduling a task, and more.' There's a 'Show All Activities' button at the bottom of the activity section.

Figure 15: Student record — Sudharshan Kuppili, CGPA 8.75, Branch CSE

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Search...

Service Home Chatter Accounts Contacts Cases Reports Dashboards Patients Appointments Placement Applications **Jobs** More

Job **Software Engineer** [New Contact](#) [Edit](#) [New Opportunity](#)

Related **Details**

Job Name	Software Engineer	Owner	Kuppli Sudharshan
Company Name	TCS		
Minimum CGPA	7.50		
Eligible Branch	CSE		
Active Backlog Limit	0		
Application Deadline	02/08/2027		
Created By	Kuppli Sudharshan, 07/08/2026, 3:16 pm	Last Modified By	Kuppli Sudharshan, 07/08/2026, 3:16 pm

Figure 16: Job record — Software Engineer, TCS, Minimum CGPA 7.50

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Search...

Service Home Chatter Accounts Contacts Cases Reports Dashboards Patients Appointments Placement Applications **Jobs** More

Job **AI Engineer – Test** [New Contact](#) [Edit](#) [New Opportunity](#)

Related **Details**

Job Name	AI Engineer – Test	Owner	Kuppli Sudharshan
Company Name	Test Company		
Minimum CGPA	9.00		
Eligible Branch	CSE		
Active Backlog Limit	0		
Application Deadline	26/08/2026		
Created By	Kuppli Sudharshan, 08/08/2026, 10:48 am	Last Modified By	Kuppli Sudharshan, 08/08/2026, 10:48 am

Figure 17: Job record — AI Engineer – Test, Minimum CGPA 9.00 (used in duplicate/eligibility testing)

Job

Java Developer Test

New Contact

Edit

New Opportunity

Related

Details

Job Name

Java Developer Test

Company Name

Test Company

Minimum CGPA

7.50

Eligible Branch

CSE

Active Backlog Limit

0

Application Deadline

31/12/2026

Created By

Kuppli Sudharshan

11/08/2026, 11:52 am

Owner

Kuppli Sudharshan

Last Modified By

Kuppli Sudharshan

11/08/2026, 11:52 am

Figure 18: Job record — Java Developer Test, Minimum CGPA 7.50 (used for the successful application, PA-0005)